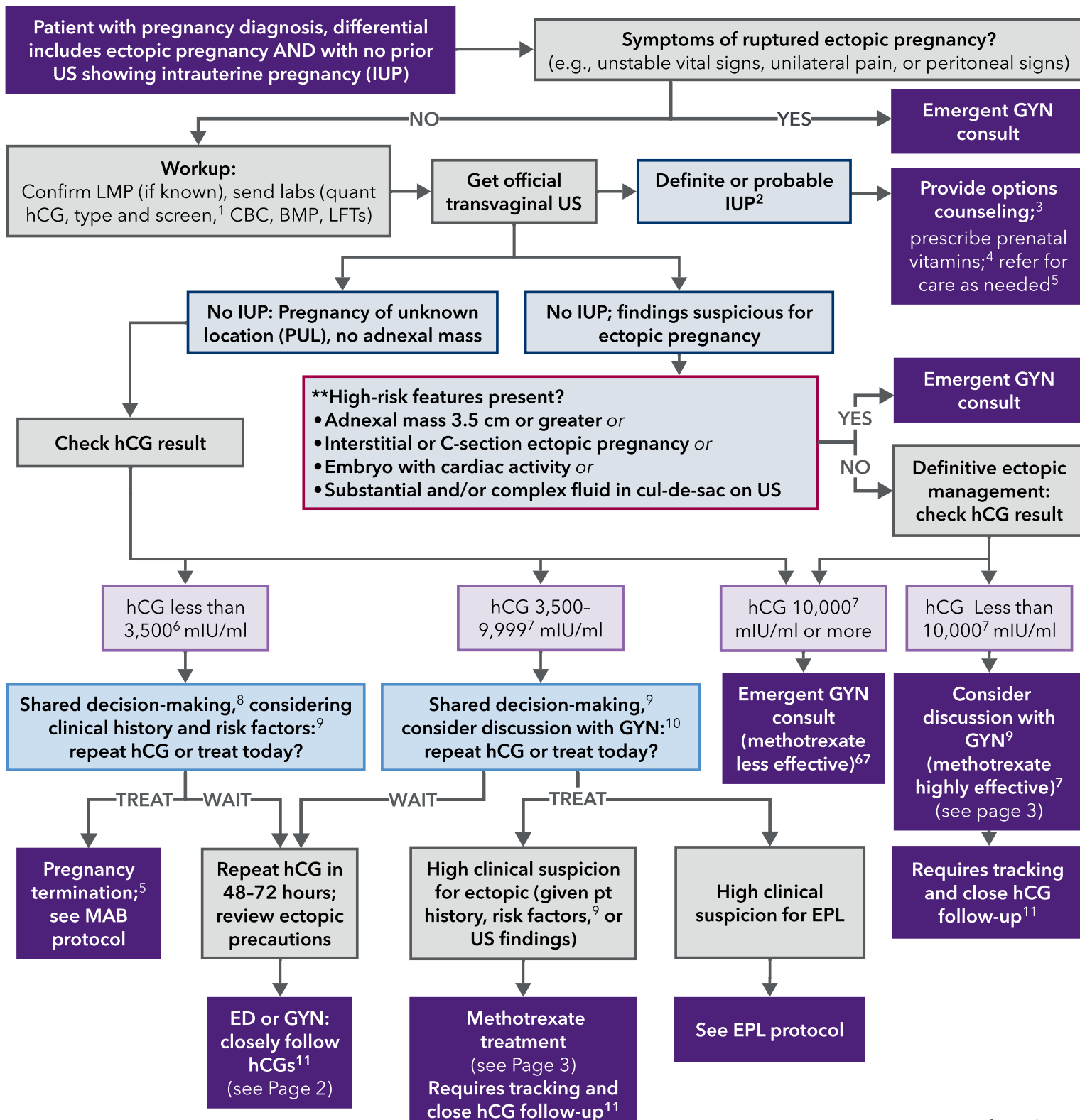


PUL & Ectopic Pregnancy: Work-Up and Management in the Emergency Department *(in states where abortion is permitted)*

Review with GYN referral/back-up to obtain mutual agreement on protocol prior to use.
PUL means pregnancy of unknown location.

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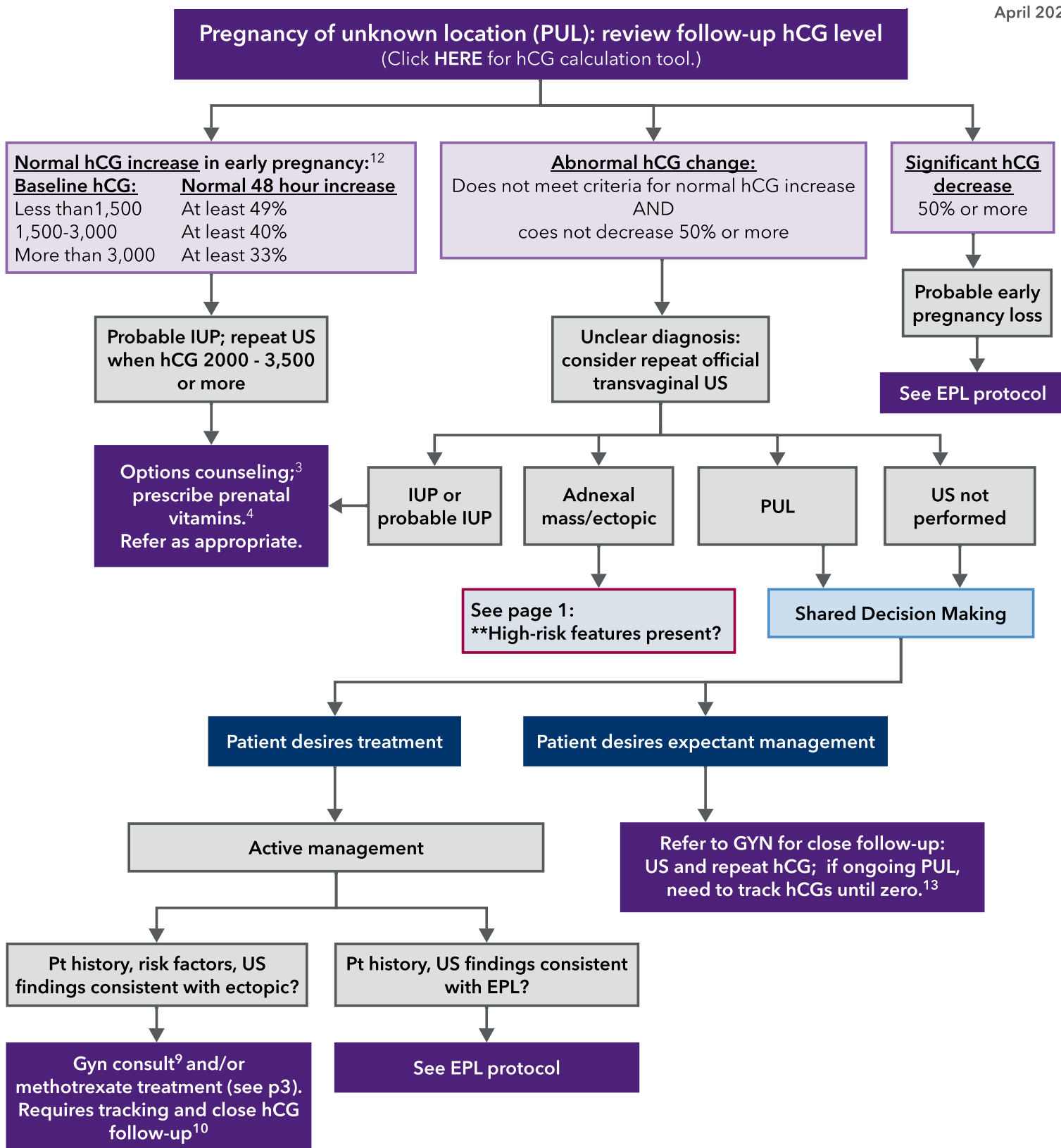


See page 4 for endnotes.

The Reproductive Health Hotline (1 844-REPROHH or 1 844-737-7644)

Free, confidential, nationwide, evidence-based hotline for health care providers. Monday–Friday 6am–5pm PT / 9am–8pm ET.

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- Must have arranged referral relationships for follow-up OR an assigned person in facility to track hCGs until zero.
- Patients must have no absolute contraindications.

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Contraindications to Methotrexate Therapy¹²

Absolute contraindications

- Ruptured ectopic pregnancy
- Hemodynamic instability
- Intrauterine pregnancy
- Evidence of immunodeficiency
- Moderate to severe anemia, leukopenia, thrombocytopenia
- Active pulmonary disease

Absolute contraindications, continued

- Active peptic ulcer disease
- Clinically important hepatic or renal dysfunction
- Breastfeeding
- Sensitivity to methotrexate
- Inability to participate in follow-up

Relative contraindications (GYN consult)

- Cardiac activity detected on US
- High initial hCG (see below for failure rates by baseline hCG)
- 4.0 cm or larger adnexal mass on TVUS
- Refusal to accept blood transfusion, if needed

Baseline hCG less than 5,000⁷ and no adnexal mass, or mass less than 3.5 cm¹⁴Baseline hCG 5,000-9,999⁷ and/or adnexal mass more than 3.5 cm;¹⁴ recommend GYN consult

Day 1 (date of injection)	Perform CBC, AST, ALT, creatinine If within normal limits, MTX IM 50 mg/m ²	Perform CBC, AST, ALT, creatinine If within normal limits, MTX IM 50 mg/m ²
Day 4	Repeat hCG. Day 1 to day 4 rise is expected.	Repeat dose of MTX IM 50 mg/m ² Repeat hCG. Day 1 to day 4 rise is expected.
Day 7	Repeat hCG. Day 4 to Day 7 levels should drop at least 15%. If less than 15% drop: repeat MTX IM 50 mg/m ² (same as above) or refer to surgery. If 15% or more drop, MTX is working. Repeat hCG weekly until zero.	Repeat hCG. If drop between Day 4 and 7 is less than 15%, refer to GYN for repeat MTX IM or surgery. If 15% or more drop, MTX is working. Repeat hCG weekly until zero.

Patient Education

- Avoid folic acid, prenatal vitamins, alcohol, NSAIDS, sexual intercourse and strenuous activity until completion of treatment.
- Mild-moderate abdominal pain, nausea and vomiting are normal.
- Go to ED if severe abdominal pain or dizziness (signs of rupture).
- Must follow hCG weekly until zero.
- Avoid pregnancy for three months (use reliable contraception).
- Stay within 30 minutes of an ED until treatment complete.

Success Rates of Single-Dose Methotrexate IM for Ectopic Pregnancy⁷

Initial hCG level	Success rate
Less than 1,000	98%
1,000 to 1,999	94%
2,000 to 4,999	96%
5,000 to 9,999	85%
10,000 or more	81%

Ectopic Risk Factors

- Previous ectopic pregnancy (10% risk with history of one ectopic, 25% risk with history of two ectopics)
- Pregnancy while intrauterine device in place (50% risk)
- Pelvic inflammatory disease or history of chlamydia or gonorrhea
- Previous fallopian tube surgery or documented fallopian tube pathology
- Infertility and history of fertility treatments
- Age over 35 years
- Cigarette smoking

Interpreting hCG Levels¹²**hCGs changes over time are variable**

- Normal rise does not guarantee normal pregnancy.
- Abnormal levels do not always indicate abnormal pregnancy.
- Consider GYN consultation in strongly desired pregnancies.
- hCGs typically plateau at 10 weeks gestation; hCG trending is not predictive in later pregnancy
- **Normal rate of hCG increase** after 48h for viable pregnancy:
 - Baseline hCG <1,500: at least 49%
 - Baseline hCG 1,500-3,000: at least 40%
 - Baseline hCG >3,000: at least 33%

Early Pregnancy Loss: normal decrease (if less, suggests ectopic)

- After 48 hours: expect at least 21% decrease
- After 4 days: 85% decrease rules out ectopic
- After 7 days: expect at least 60% decrease; 95% rules out ectopic
- In the setting of an IUP of uncertain viability, hCG rise less than 11% is diagnostic of nonviable pregnancy.¹⁵

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For additional Access Bridge resources, visit bridgetotreatment.org/access-bridge/.

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Ectopic Pregnancy: Work-Up and Management in the Emergency Department *(in states where abortion is permitted)*

ENDNOTES:

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- 1 Type and screen for potential cross match. Rh testing and RhoGAM administration are not recommended before 12 weeks' gestation for EPL, medical abortion or procedural abortion ([ACOG 2024, SFP 2022](#)). RhoGAM recommended for Rh negative patients with ectopics. Follow institutional guidance.
- 2 Probable IUP means intrauterine fluid (sometimes labeled as gestational sac) without a yolk sac; a retrospective study found the incidence of ectopic pregnancy in patients with intrauterine fluid collections was very low; follow-up ultrasound in one week can be offered. [Flynn 2023](#).
- 3 All-Options Talkline 1-888-493-0092; see [options counseling guide](#).
- 4 Folic acid prevents birth defects if taken early in pregnancy (see [CDC](#)).
- 5 Options for pregnancy termination can be found on [INeedAnA.com](#).
- 6 The discriminatory zone is the hCG level at which a gestational sac (GS) should be visible on transvaginal US. hCG level of 3,500 is the threshold used by ACOG. ([Connolly 2013](#)).
- 7 If methotrexate used, failure rate based on pre-treatment hCG: 3.7% if <5,000; 15% if 5,000 - 10,000; 19% if $\geq 10,000$ ([Menon 2007](#)).
- 8 Shared decision-making is appropriate for patient with a PUL, considering desirability of pregnancy. Options include: diagnostic aspiration or mifepristone and misoprostol, methotrexate, or expectant management. If intrauterine fluid (possible gestational sac) is present with no adnexal mass, mifepristone and misoprostol or aspiration may be most appropriate if treatment desired. [Flynn 2023](#).
- 9 Tailor treatment and follow-up to patient's risk factors for ectopic pregnancy: 50% if IUD in place at time of pregnancy, 10% if history of prior ectopic, 10-15% with prior tubal surgery. Other risk factors: estimated GA by LMP over 42 days if PUL on ultrasound, history of PID, chlamydia or gonorrhea. [ACOG 2018](#).
- 10 If available and clinically appropriate, diagnostic aspiration to rule out EPL may be done prior to methotrexate treatment. [Flynn 2023](#).
- 11 Requires responsible party to track patient and call if no show for follow-labs (by ED case management or confirmed "warm hand-off" to outpatient clinic, GYN, or other referral partner).
- 12 From [ACOG Tubal Ectopic Pregnancy](#).
- 13 In setting of PUL and suspected ectopic pregnancy, need to follow hCGs until zero. [ACOG 2018](#). Use shared decision-making with patient weighing relative risks, costs, and availability of follow-up. 7% of PULs end up being ectopic pregnancies. [Park 2023](#).
- 14 Ectopic pregnancies are more likely to rupture over 3.5cm; recommend GYN consult prior to treatment. [Barash 2023](#).
- 15 In the setting of an IUP of uncertain viability, hCG rise less than 11% is diagnostic of nonviable pregnancy. ([Society of Family Planning Medication Management of Early Pregnancy Loss](#))

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